An orbit is continuous falling

antocciphysics | TEACHER KEY | CPA

Learning target: Explain a circular orbit with gravity and sideways motion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 10; 3 course days later.*
24 m from the base in the launch direction.

02. Explain a worked example (worked)
Tangent velocity changes direction; inward force points to the central body throughout.

03. Look and explain (model)
Sideways motion carries it around while gravity curves path; center pull, tangent velocity.

04. Build the solution (practice)
3 orbits; 5 m/s.

05. Use the idea (reason)
Gravity supplies the inward net force; continued tangent motion does not need forward thrust. Drag/orbital changes would require corrections.

06. Choose a method (mixed) *Review: Unit 2, Day 1; 12 course days later.*
Changing heading or a curved sequence of positions shows direction changes; velocity includes direction.

07. Exit check (exit)
Yes; inward acceleration changes direction.

08. STRETCH: Apply it another way (stretch)
The 4 m path has twice the circumference and period. Same speed at both radii is not generally a same-body gravitational circular-orbit pair.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 10; 3 course days later.*
t=2 s; range 10 m; vertical velocity 20 m/s downward.

02. Explain a worked example (worked)
Tangent velocity changes direction; inward force points to the central body throughout.

03. Represent and interpret (model)
Sideways motion carries it around while gravity curves path; center pull, tangent velocity.

04. Solve and justify (practice)
v 5 m/s; T=2π16/5=20.1 s; a 25/16=1.56 m/s².

05. Analyze the reasoning (reason)
Gravity supplies the inward net force; continued tangent motion does not need forward thrust. Drag/orbital changes would require corrections.

06. Choose a method (mixed) *Review: Unit 2, Day 1; 12 course days later.*
Direction changes but speed remains 6 m/s. Instantaneous perpendicular force does no work; velocity is not constant.

07. Exit check (exit)
Speed halves; period increases 8 fold because circumference 4×and speed .5×.

08. STRETCH: Apply it another way (stretch)
Speed becomes 1/3, period 27 times, acceleration 1/81 of the originals.